ALEX HOVER

Senior .NET Engineer

(845) 377-1325 | alex.hover@gmail.com | linkedin.com/in/alex-hover-9b08a4338/

Columbia, South Carolina

EXPERIENCE

Senior .NET Engineer

Sentry Pay

10/2022 - 11/2024 | Tampa, FL

- Built a **microservice-based**, event-driven backend with Domain-Driven Design, **.NET**, **Angular**, and **Azure** to drive performance and scalability for the Sentry Pay financial trading platform.
- Built a high-performance backend with **ASP.NET Core**, Web API, and **EF Core** to deliver reliable APIs, manage messaging, and optimize data access for Sentry Pay.
- Optimized **RDBMS** and NoSQL systems (**MS SQL**, Azure SQL) and implemented caching with **Redis** to support scalable growth and high-traffic demands for Sentry Pay.
- Developed **RESTful APIs**, **GraphQL**, and **gRPC** services, and implemented an event-driven system using **Kafka**, **RabbitMQ** for real-time data processing to support the scalability and integration of Sentry Pay's platform.
- Streamlined Azure cloud infrastructure using **Git**, **Docker**, **Kubernetes**, **Terraform**, Azure DevOps, and AKS to automate deployments, manage version control, and enhance scalability for Sentry Pay.
- Enhanced productivity and collaboration by **integrating third-party services** like **Stripe**, **PayPal**, Google, and Facebook for secure, seamless transactions, improving team efficiency and user experience on Sentry Pay's platform.
- Implemented a **fraud detection** model using **Python** and **machine learning** to enhance fraud prevention at Sentry Pay.
- Implemented strong security between microservices and external communications using **OAuth 2.0**, **JWT**, and secure API authentication to ensure data privacy and integrity across Sentry Pay's platform.
- Led QA efforts using **xUnit**, **Jasmine**, and Karma for testing, while managing tasks in **Atlassian Jira** to ensure platform reliability through collaborative code reviews at Sentry Pay.

Full Stack Developer

Apple

12/2020 - 09/2022 | South Carolina

- Spearheaded the development of a scalable, microservices-driven architecture with event-driven design, Domain-Driven Design (DDD), **.NET**, **Angular**, and **AWS**.
- Built core backend systems using ASP.NET Core, .NET Web API, and Python, ensuring system robustness and efficiency.
- Enhanced communication architecture with Kafka, RabbitMQ, and GraphQL, gRPC services, enabling real-time data processing and seamless inter-service communication.
- Optimized databases and caching with **MongoDB**, AWS DynamoDB, **Redis**, and ElastiCache to efficiently handle transaction demands, significantly improving data access speed and overall system performance.
- Implemented DevOps on AWS cloud with Git, Docker, Kubernetes (EKS), Terraform, and Lambda, using EC2, S3, and other AWS services to improve scalability and streamline deployments.
- Built a high-performance **micro-frontend** with **Angular**, TypeScript, NgRx, and RxJs, utilizing lazy loading and optimized state management to improve load times.
- Led QA efforts with xUnit, Jasmine, Karma, AWS, Agile methodologies, and Atlassian Jira to enhance reliability and reduce bugs.

SUMMARY

Senior .NET Engineer with **7 years** of experience designing and implementing high-performance, scalable, and complex solutions across technology, cloud computing, and web applications.

Expert in **.NET** and **C#**, with extensive skills in architecting **microservices-based** and **cloud-native systems**. Proficient in ASP.NET Core, Entity Framework, RESTful APIs, GraphQL, gRPC, SQL/NoSQL databases with hands-on experience in **Azure** and **AWS** for cloud deployment and scalability.

Highly collaborative, adept at technical specification writing, and skilled in **cross-functional team leadership**, **agile methodologies**, and effective communication, bringing a strategic and results-oriented approach to every project.

SKILLS

Programming Languages & Frameworks

C# | **.NET** | **ASP.NET Core** | **JavaScript** | **TypeScript** | **Python** | **Angular** | **RxJS** | **NgRx** | **Node.js**

Web & Application Development

OOP | **Data Structures** | **RESTful APIs** | **Microservices Architecture** | **Swagger** | **MVC Framework** | **Web Server** | **HTML5** | **CSS3** | **Bootstrap**

Database Management

Relational Databases (RDBMS) | **MS SQL** | **ADO.NET** | **Entity Framework** | **MySQL** | **PostgreSQL** | **NoSQL Database** | **T-SQL** | **MongoDB** | **Redis** | **Apache Kafka** | **RabbitMQ**

EXPERIENCE

Backend Engineer

Microsoft

09/2018 - 12/2020 South Carolina

- Built a high-performance backend with ASP.NET Core, .NET Web API, and Entity Framework, providing reliable APIs and efficient data access.
- Designed and implemented an event-driven architecture with Kafka, RabbitMQ, microservices, RESTful APIs, and GraphQL to enable real-time data processing and support platform growth.
- Optimized data architecture with MS SQL, Azure SQL, and Redis, improving query speeds, data retrieval efficiency, and response times for high-traffic demands.
- Supported DevOps optimization with Git, Docker, Kubernetes, AKS, and Terraform, enhancing deployment speed and scalability.
- Developed high-performance frontends using Angular, NgRx, Tailwind CSS, and Apollo GraphQL, leveraging lazy loading and optimized state management to improve load times.

.NET Developer

ScholarJet

08/2017 - 09/2018 Boston, MA

- Contributed to the Education project by designing and incorporating MVC patterns and object-oriented principles to optimize transaction processing speed, lower maintenance costs, and enable scalable growth.
- Built a high-performance backend using ASP.NET, ADO.NET, and MS SQL to deliver reliable APIs and efficient data access, achieving faster response times and increased system reliability.
- Developed RESTful APIs with JWT and OAuth to streamline data flow and enhance interoperability across systems, achieving faster response times and seamless integration with third-party applications.
- Developed and supported QA efforts with unit, integration, and automated testing using xUnit, NUnit, MSTest, Selenium, and Azure DevOps Test Plans, applying SOLID Principles to reduce bugs and strengthen platform reliability.
- Migrated frontend components to Angular, improving modularity, reducing load times, and enhancing integration with backend services.

EDUCATION

Master of Computer Applications in Computer Science

Harvard University

11/2016 - 08/2017 Cambridge, MA

Bachelor of Science in Computer Science

Harvard University

09/2012 - 10/2016 Cambridge, MA

SKILLS

Software Development & Methodologies

Technical Leadership · Scrum · Kanban · Agile Methodology · SDLC

Cloud Computing & DevOps

Microsoft Azure · Azure App Service · Cosmos DB · Functions · Blob Storage · SQL Database · AKS · Azure DevOps · AWS · S3 · EC2 · RDS · DynamoDB · API Gateway · GCP · Docker · Jenkins · Kubernetes · CI/CD Pipelines · Nginx · Terraform · GitHub Actions · TFS · Git · GitHub · GitLab

Testing & Quality Assurance

Quality Assurance (QA) · TDD · xUnit · Unit Testing · NUnit · Debugging · SOLID Principles

Tools & Productivity

Visual Studio · VS Code · Microsoft 365 · Atlassian JIRA · Stripe · PayPal